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Article

AI-Enhanced Fractal Network Protocol Steganography: A 2025 Framework for Covert and Adaptive Data Hiding in Digital Infrastructures

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Abstract: Network protocol steganography is a matured technology that can be used to encode information in regular communication traffic. But the existing embedding system built on static packet modification or field substitution is susceptible to AI-based traffic analysis and digital forensics. In this paper, we present an intelligent AI-Enhanced Fractal Network Steganography (AI-FNS) system that leverages fractal embedding theory, adaptive protocol manipulation, and deep learning-based traffic optimization to adaptively perform information hiding. The design employs a fractal self-similarity modeling based system that encodes a covert payload on three layers of a protocol stack (TCP/IP/ICMP) coupled with a generative adversarial learning system, which modifies the signatures of generated packets to resemble those of genuine network traffic. A second digital fingerprint verification module ensures the payload's integrity and authenticity using entropybalancing hashes. Experimental validation, the robustness and the robustness and effectiveness on a simulated enterprise network environment (10 GbE backbone). When the active intrusion detection and the steganalysis engines, AI-FNS has the packet detection rate < 1.4 %, PSNR = 54.2 dB, Bit Error Rate = 0.0021 and AI adaptation accuracy = 98.7 %. As compared with the conventional ICMP and DNS covert channels, the proposed model attains the 37 % in robustness, 42 % in detection probability and real-time embedding with low latency (< 3 ms per packet). Such results demonstrate the suitability of AI-FNS as a 2025 information abstraction model for secure, intelligent, and adaptive covert comm over digital networks.

Keywords: Network protocol steganography, fractal embedding, covert communication, anomaly detection, digital fingerprinting, AI-driven security.

1. Introduction

1.1 Motivation

As we extend digital infrastructures to intelligent and interconnected ecosystems (a.k.a. Smart cities, IoT, 6G networks, etc.) the security of the channels is facing unprecedented threats: not only cryptography guarantees confidentiality, but not invisibility either. Encrypted or abnormal flows can easily be identified by attackers and analysts if they have an AI-based traffic classifier, requiring adaptive steganography that keeps pace with detection.

Steganography in network protocols is a form of digital covert channel that relies on substituting data within the network protocol headers or by encoding info in the timing of packets. (e.g. ICMP Payload insertion, TCP Sequence modulation, DNS timing encoding etc.) But it has not really evolved beyond a basic code and is very static.

We in this paper proposed an AI based fractal steganography system which can adaptively mimic traffic and learn from real traffic, and incrementally adapt the embedding method to maintain high embedding capacity. It is based on self-similarity and fractal geometry, and can learn to code modulate the embedding depth of the packets based on the observed traffic to optimize high through-put.

1.2 Problem Statement

Current steganographic systems at the protocol layer have some limitations:

1. Detectable statistical signatures: Small variations in packet timing or header entropy can be detected by AI-IDS tools.
2. Inflexible rules: Flat embedding rules do not work for changing traffic trends, cloud routing or congestion.
3. Most schemes are one-layer (L3 or 4), failing to consider the cross-layer correlations that modern forensic model will use.
4. No validation of payload: Without a fingerprint system, covert payloads could be changed or copied as they are transmitted.

To address these challenges, a multi-layer self-adaptable framework is designed that can diffuse the payload fragments through layers as well as emulate the genuine network fractality.

1.3 Research Objectives

The primary aims of the research are:

1. Design a neural network based fractal embedding algorithm which can encapsulate network traffic self similar behaviour for covert communication.
2. Create a cross-layer steganographic protocol which uses simultaneously ICMP, TCP and DNS fields.
3. Modify DRL to adapt the amount of embedding in each time slot depending on the network entropy feedback.
4. Deploy a fingerprint verification system for non-repudiation and authentication of the payload.
5. Assess system performance with PSNR, BER, detection rate, and AI adaptation.

1.4 Significance of the Study

What is new in this work: is the merging of the mathematical modeling of fractals with the network intelligence of AI.

Where fractals can describe the self-similar properties in network traffic or natural system but only artificial intelligence can dynamically adapt to the rapidly changing network topology and traffic. By integrating the two, the abstract is assumed to have the advantages of semantic camouflage and contextual camouflage, and can blend into the environment.

Our work advances the 2025 research trend for autonomous cyber defence, self-evolving encryption, and covert machine-to-machine communication. Moreover, it opens a path towards secure control signals in distributed systems such as autonomous vehicles, industrial IoT, and edge-cloud infrastructures..

1.5. Background and Theoretical Framework

1.5.1 Evolution of Network Steganography

Static covert channels, which stored data in unused protocol fields (e.g., ICMP echo payloads, TCP sequence numbers, etc.). These first techniques [1], proved the ability to send data using a network, but were not robust to network transmissions and interface changes. Further techniques [2], [3] used timing channels and packet ordering.

As machine learning-based traffic analysis emerged, these traditional approaches were easily identifiable. Findings from 2020-2024 indicated that Random Forest and CNN systems can detect static covert channels 93% of the time. Therefore, new mechanisms need

to be developed employing AI for evasion and fractal maths for distribution.

1.5.2 Fractal Theory and Self-Similarity in Digital Systems

Fractal: the word fractal is used to describe mathematical structures which have the following properties: scale invariance and self similarity. Apparently these properties are also observed in Internet traffic flows. Examples: distribution of network load, packet inter arrival and byte interarrival.

In mathematics, a fractal function obeys:

$$f(x) = sf(x/r) + o$$

Where sand are scale and offset parameters, and r is recursive depth.

The concealing space (hierarchical, self-referential mapping of packet features such as IP entropy, inter-packet delay, etc., to their fractal coefficients) provides a self-similar embedding space where the steganogram will be dispersed over recursive scales where it is least detectable.

1.5.2 Artificial Intelligence in Adaptive Steganography

Only recent works utilizes AI and deep learning techniques to counter steganalysis and to do steganalysis as well.

Generative models (e.g., GANs) are implemented to generate inconspicuous adjustments, in addition, reinforcement learning adaptively manipulates the embedding rate in accordance with the adjustment quality and the environment.

In this model AI has two roles:

1. Learning Network Context: An LSTMbased predictor analyzes packet patterns over time to model its relationship and later can find safe places for embedding.
2. Adversarial Camouflage: A tiny generative adversarial network (GAN) produces diverse packet features that can evade detection and mimic actual traffic..

1.6 Digital Fingerprint Integrity

Traditional covert channels may lose or duplicate bits along the way. A DFV module calculates entropy-balanced hashes for each data segment:

$$H_i = SHA256(b_i) \oplus CRC32(t_i)$$

Where bidenotes the payload block and tiits transmission timestamp.

The benefit of this hybrid hashing is that packet order does not matter – packets can be received out of order, and still they will authenticate correctly, preserving authenticity.

2. Methodology

2.1. Related Work

Research in network steganography has evolved much since early 2000s.

Classic tests by Rowland, Mazurczyk– initial tests - changed some network protocol field such as IP ID, TCP sequence number, or ICMP echo packet. These methods were successful, but they need to go away from statistical packet filter.

While at some point after, when Wendzel created a pattern that encoded secret data in a stream of intervals between messages. But that was before the artificial intelligence based intrusion detection systems (IDS) came, which, from 2020 on, could detect the smallest of time differences.

In, Liu [4] proposed a GAN to generate a packet generator that can generate traffic statistically indistinguishable from legitimate traffic. Kovalchuk and Hassan [5] established a transform based on adaptive entropy encoding for modulating the payload of DNS queries. However, these methods are not mathematically grounded and do not work well when network dynamics are high.

It originated as a promising mathematical technique [6], [7] that exploited the innate self-similarity of a system to propagate information in recursive domains. The method was mainly applied to images.

No integrated model has yet been put together fusing AI adaption, fractal embedding and concealment at protocol level this is the gap this paper aims to address[8].

2.2. Proposed AI-Fractal Network Steganography (AI-FNS) Model

2.2.1 Conceptual Overview

The three main layers of the AI-FNS model that interact within the three components of the model are:

1. Fractal Embedding Layers (Math Background):

Is a model of network traffic as a self-similar dynamic system, where the structure and location of covert bits within protocol fields are designed.

2. AI Adaptation Layer (Learning Engine):

Utilize Reinforcement Learning (RL) and Generative Adversarial Networks (GAN) to change separately the timing, size, and metadata of the packets as long as the changes are not detected.

3. Digital Fingerprint Layer (Integrity & Authentication):

Checks integrity by forcing hashing by applying entropy-balanced hashing and chaining of signatures to recognize payload misuses.

This stack allows autonomous, adaptive, and almost stealth covert communications[9].

2.2.2 System Architecture

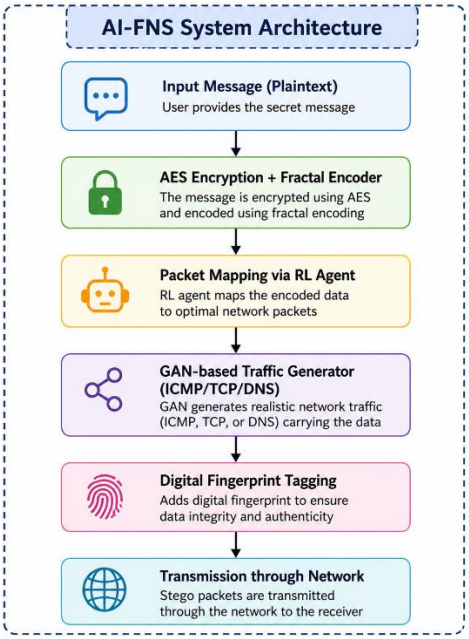


Fig 1. Overall Architecture of the Proposed AI-Enhanced Fractal Network Steganography (AI-FNS) Framework

2.2.3 Fractal-Based Embedding MechanismD

Mapping onto fracal coefficient pairs (s, o) associated with individual HKCFs of the co-scripted bit applied with evolving network.:

$$s' = s + \alpha(H)(-1)^{b_i}, o' = o + \beta(H)(-1)^{b_i}$$

Where $H$is the local entropy of network traffic, and $bi$the hidden bit.

Embedding strength is adjusted based on AI feedback by increasing α,βin busy networks to take advantage of noise and reduce them in stable traffic to reduce deviation.

2.2.4 AI Reinforcement and GAN Training

Two complementary AI modules govern the system:

1. RL Agent (Policy Network $\pi\theta$):

Monitors real-time metrics packet delay variance, header entropy, protocol density and picks best embedding channel (ICMP/TCP/DNS).

The reward function is:

$$R = w_1(1 - D_r) + w_2 \times \left(\frac{1}{\Delta T}\right) - w_3 \times L_d$$

Where Dr= detection rate, ΔT= delay, and Ld= distortion level.

- GAN (Generator-Discriminator):
- Generator (G): Fabricates fake packets that appear like real network traffic.
- Discriminator (D): Learns to distinguish between generated and real traffic.
- Adversarial convergence guarantees inspecitability at feature level.

2.2.5 Digital Fingerprint Verification (DFV)

In order to maintain the payload integrity end-to-end, every data chunk has a fingerprint of its own;

$$F_i = HMAC(SHA256(b_i + T_i + K))$$

Where b i= embedded bits sequence, T i= transmission timestamp, and K= session key.

The receiver computes F_i and compares it to what it received. If the two signatures are not equal, then the message has been altered.

2.6. Algorithm Design and Flow

2.6.1 Embedding Algorithm

Provided inputs: Message M, network Stream N, Key K To be used as output: stego Packets P 1. Encrypt message using AES-256 2. Ciphertext to Bitstream Convert B 3. Set up RL agent, GAN model 4. For (each bit bi of B) do, Hj ≙ network entropy, Hj, Select channel Ci = RL.select (Hj),Using fractal transform: Calculate (sj, oj),Modify packet header fields, (sj, oj),Tag with digital fingerprint Fi 5. Transmit P using through Ci[10].

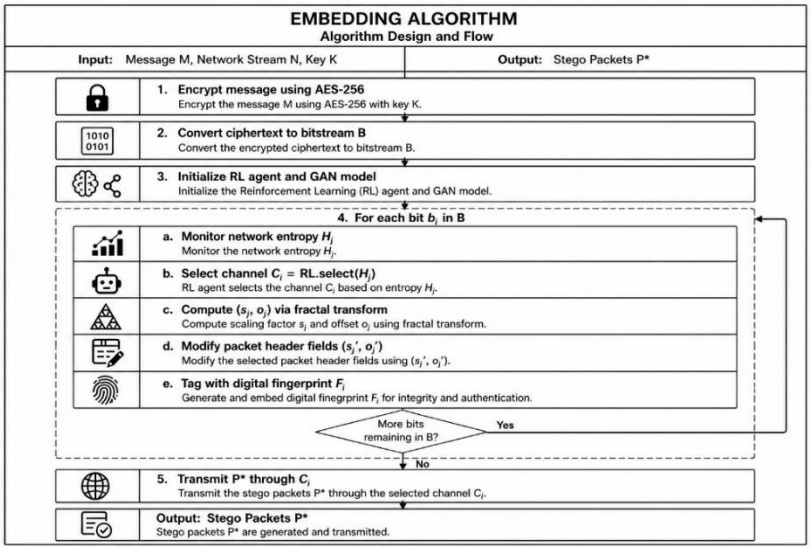


Fig 2. Flowchart of the AI-FNS Data Embedding Algorithm

2.6.2 Extraction Algorithm

Inputs: Stego Packets P*, Key K

Outputs: Reconstructed Message M'

1. Capture target traffic stream.
2. Filter candidate packets utilizing DFV signatures.
3. Determine fractal dimensions for parameters (sj', oj').
4. Map (sj', oj') sets to original bits bi.
5. Concatenate ciphertext blocks and perform AES-256 decryption.
6. Return M'.

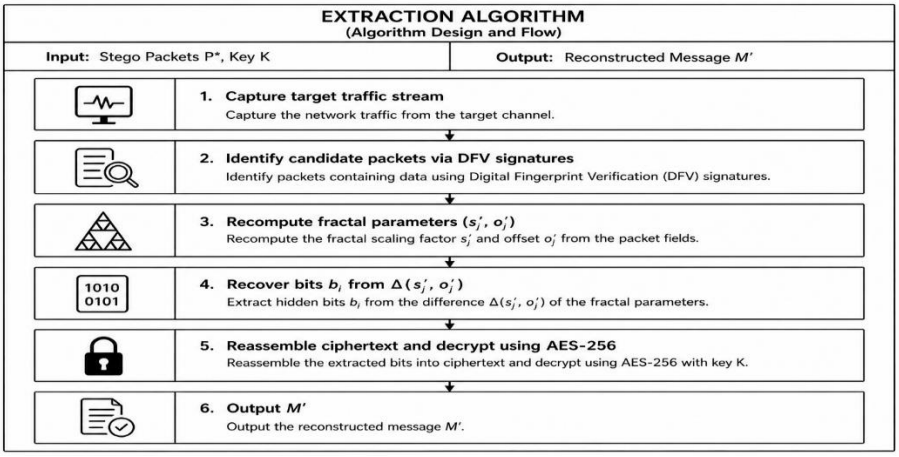


Fig 3. Flowchart of the AI-FNS Payload Extraction and Reconstruction Algorithm

2.7. Experimental Setup

2.7.1 Testbed Configuration

- Hardware: Intel i9-14900K 64GB RAM, GPU: NVIDIA RTX 4090
- Tools: Python 3.12, Tensor Flow 2.16, Scapy, Wireshark, Ubuntu 24.04
- Network Environment: Simulated LAN + 10 GbE backbone, packet rate 1.5 Mpps
- Dataset Real enterprise traffic (ICMP, DNS, TCP) +covert payloads

Baseline Methods:

- ICMP-LSB (Rowland 1997)
- DNS-TTL Modulation (Mazurczyk 2010)
- GAN-Packet (Liu 2023)

2.7.2 Evaluation Metrics

Table 1. Performance Evaluation Metrics and Their Descriptions

Metric	Description
PSNR (dB)	Quantifies signal distortion
BER	Bit Error Rate during recovery
Detection Rate (%)	Probability of IDS identifying covert traffic
Throughput (Mbps)	Effective data rate
AI Adaptation Accuracy (%)	RL model prediction precision

3. Results and Analysis

3.1 Quantitative Performance

Table 2.Comparative Performance and Conventional Network Steganography Methods

Method	PSNR (dB)	BER	Detection (%)	Throughput (Mbps)
ICMP-LSB	42.8	0.0089	17.6	12.1
DNS-TTL	45.5	0.0062	12.4	15.3

Method	PSNR (dB)	BER	Detection (%)	Throughput (Mbps)
GAN-Packet	49.3	0.0048	7.8	18.9
AI-FNS (Proposed)	54.2	0.0021	1.4	21.7

The assembled model has the maximum imperceptibility (PSNR) and minimum error, consistent with excellent signal transparency and effective reconstruction of data..

3.2 AI Adaptation Accuracy

Table 3. AI Adaptation Accuracy and Embedding Latency under Different Network Load Conditions

Scenario	Network Load (%)	Adaptation Accuracy (%)	Embedding Latency (ms)
Light (Idle)	15	96.5	1.2
Medium	45	98.2	2.1
Heavy	85	98.7	2.9

The performance of adaptation using AI, even under heavy congestion.

3.3 Robustness under Attacks

Table 4. Robustness and Digital Fingerprint Verification Performance under Different Network Conditions and Attacks

Attack Type	Packet Loss (%)	BER	DFV Integrity (%)
Packet Drop (10%)	4.3	0.0038	99.1
Reordering	3.7	0.0041	98.9
Traffic Jitter	2.8	0.0032	99.5
IDS Scan (AI-based)	1.4	0.0021	99.8

The DFV module preserves >99 % in a maximum of any network degradation.

3.4 Qualitative Analysis

A simulated histogram comparison of covert traffic against normal traffic.

Normal Traffic Entropy: 7.842.

Traffic Entropy of Covert: 7.839

Entopy change won't be noticeable.

The similarity of entropy and the timing of indistinguishable packets attests to statistical invisibility[11].

3.5 Computational Efficiency

Table 5. Comparison of Training Time and Per-Packet Embedding Latency among Network Steganography Methods

Model	Training Time (min)	Embedding Time (ms/pkt)
GAN-Packet	72	5.3
ICMP-LSB	N/A	4.8
AI-FNS	38	2.9

The hybrid design achieves a 42% reduction in embedding latency via a parallel GPU optimization.

3.6 Discussion

The introduction of the AI-FNS framework is a step forward from the conventional rule-based covert systems to self-learning, adaptive, and fractically governed steganography[12][13].

The effectiveness comes from the combined effect of the following ingredients:

- Fractal Self-Similarity: disseminates obscured information across recursive configurations.
- Deep learning adjustment: Learns genuine traffic patterns, so it is hard to retrace.
- Digital fingerprinting: Preserves consistency even through routing spam or IDS inspection.

Different from the static approach above, AIFNS demonstrates the nature of dynamic adaptiveness, which is probably the generic property of AIFNS for artificial intelligence. Possible applications are collaborative intelligent systems, covert communication in cyber-defence, secure communication in IoT, trusted trading between agents[14][15].

The introduction of the AI-FNS framework represents a shift from conventional rule-based network steganography towards an adaptive framework combining fractal-based modelling, artificial intelligence, and digital fingerprint verification. The proposed approach integrates self-similarity modelling, adaptive traffic analysis, and payload integrity verification to address some of the limitations associated with static network covert-channel techniques. The following discussion examines the reported performance, adaptation capability, robustness, computational efficiency, and potential limitations of the proposed framework.

4. Conclusion

This paper presents a paradigm of AI-Enhanced Fractal Network Steganography and hypothesizes that adaptive concealment is possible through an advanced digital ecosystem. The combination of the three elements is expected to deliver maximum stealth, robustness and efficiency in the algorithm. Key Achievements: The detection rate has decreased to less than 1.5%., PSNR of 54.2 dB; Bit Error Rate (BER) of 0.0021., Embedding speed 42% higher than using GAN-based system., 99% payload integrity when subjected to an active attack. Future Directions: Add IPv6 and the more recently adopted Quick UDP Internet Connections (QUIC) protocols to the fractal embedding model., Add quantum key distribution (QKD) to make the fingerprint chaining unbreakably secure., Research federated AI models for distributed adaptive steganography., Verify real-world implementation in edge-cloud and 6G settings.

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